

Report R5 — Dissipative rules: recurrent classes, attractors, and protected coherence

Krylov sector analysis of quantum cellular automata (Tier 1a/1b)

QCA Sector Analysis project (Tier 1)

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Abstract

We extend the exact pipeline to the 240 dissipative quantum cellular automata (rules containing a reset channel D or E). The one-cycle channel is now genuinely non-unitary: the directed transition graph has *transient* strongly connected components flowing into *recurrent classes* (attractors). We sweep all dissipative rules (both boundary conventions, $N = 6 \dots 13$), census their attractor structure, and classify each recurrent class by the peripheral spectrum of its exact restricted superoperator (Tier 1b) as a classical pointer state (*mixing*), a *limit cycle*, or a *coherence-carrying* attractor. Finally we quantify, exactly at small N , the *protected-coherence gap* — the amount by which the full channel’s fixed-point space exceeds the classical pointer count — which the transition graph provably under-resolves for weakly dissipative rules.

1 Dissipative graph structure

For a dissipative rule the channel $\Phi_C(\rho) = \sum_{\mu} K_{\mu} \rho K_{\mu}^{\dagger}$ has ≥ 2 Kraus operators. The exact engine propagates every Kraus branch (never merging incoherent alternatives) and records the directed transition graph $M_{yx} = \sum_{\mu} |\langle y | K_{\mu} | x \rangle|^2$. Its terminal strongly connected components are the recurrent classes (attractors); non-terminal components are transient. We record, per (rule, N , bc): the number and sizes of recurrent classes, the transient depth (longest condensation-DAG path), and the basin structure (states reaching a single vs. several attractors). The engine and its dissipative branch machinery are validated against an independent dense-channel oracle (528/528 support checks; per-site Kraus completeness $A_0^{\dagger} A_0 + A_1^{\dagger} A_1 = \mathbb{K}$).

2 Census of attractor structure

Table 1 classifies every dissipative rule by the structure at the largest computed N : a unique classical fixed point, several point attractors (multistability, signalling a strong basis-diagonal conserved charge), or one or more extended (> 1 -state) attractors. These are properties of the *dephased* chain M ; §4 tests when that coincides with the coherent channel’s attractors (it does for 202 of the 240 rules).

Two ground-truth items (context §8) are reproduced: rule 22 (IVVD) on **pbc** at even N has exactly three point attractors (the vacuum and the two Néel states); rules 28/29 illustrate the boundary/wall-transparency contrast.

3 Tier 1b: attractor types

For each recurrent class R we build the *exact* restricted superoperator Φ_R on $\text{span}(R) \otimes \text{span}(R)^*$ from the labelled Kraus branches, and read its peripheral spectrum (geometric multiplicities,

attractor structure	obc0	pbc
unique fixed point	66	36
multistable (point attractors)	77	75
single extended attractor	63	47
multiple extended attractors	34	82
total dissipative rules	240	240

Table 1: Dissipative rules by attractor structure and boundary convention.

robust to the non-normal transient Jordan structure). A class is *mixing* (a unique steady state; a classical pointer/dark state when $|R| = 1$), a *limit cycle* (peripheral roots of unity, basis-diagonal eigenoperators), or *coherence-carrying* (a decoherence-free subsystem of dimension $d_k > 1$). Table 2 surveys representative rules.

W	tuple	#recur	attractor kinds	max d_k
0	DDDD	1	1×mixing	1
1	VDDD	1	1×unresolved	1
22	IVVD	3	3×mixing	1
28	IIVD	11	3×mixing, 8×coherent	3
29	EIVD	10	2×mixing, 8×coherent	3
50	DVVV	1	1×mixing	1
76	IIID	131	131×mixing	1
178	DVVE	2	2×mixing	1
200	DIID	90	90×mixing	1
232	DIIE	46	46×mixing	1

Table 2: Tier-1b attractor classification (pbc, $N = 8$). Rule 28 carries coherence-protecting extended attractors ($d_k > 1$); most others relax to classical pointer states.

4 Strong vs. weak symmetry: what each stage can see

A *strong* symmetry commutes with every Kraus operator individually; a *weak* symmetry only makes the channel covariant (context §5). For unitary rules there is a single Kraus operator and the two coincide. This pipeline does **not** detect symmetry *operators* (no commutant of $\{K_\mu\}$, no charge null-space — that is Tier 2); it separates the two *operationally*, because the two stages of the pipeline are sensitive to different things:

- **Tier 1a (graph).** The graph encodes $M_{yx} = \sum_\mu |\langle y|K_\mu|x\rangle|^2$, i.e. the chain obtained by *dephasing in the computational basis after every cycle*. Only basis-diagonal *strong* charges fragment it; their signature is the number of recurrent classes (multistability, Table 1). This is the only symmetry information the graph carries.
- **Tier 1b (channel).** The exact Cesaro rank $\dim \text{Fix}(\Phi)$ (geometric multiplicity of eigenvalue 1 of the full 4^N superoperator) sees the coherent fixed-point algebra. Weak symmetries and protected coherences appear here and *only* here.

The gap $\dim \text{Fix}(\Phi) - \sum_R \dim \text{Fix}(\Phi_R) \geq 0$ measures what the graph misses (Table 4).

4.1 Caveat: the graph is the dephased chain, and can misplace attractors

Φ ’s populations are *not* determined by M — coherences feed back into populations — so “transient for M ” does not imply “transient for Φ ”. We therefore test faithfulness exactly

(Table 3) by asking whether $\text{Fix}(\Phi)$ is contained in the block spanned by the graph’s recurrent states, and whether $\Phi^t(\mathbb{K}/d)$ actually flows onto them. Two regimes occur, and the distinction matters:

- **Graph-faithful** (202 of 240 rules at $N = 4$, pbc): e.g. rule 200 — $\text{Fix}(\Phi)$ lies entirely in the recurrent block and $\Phi^{400}(\mathbb{K}/d)$ converges exactly onto the ten pointer states. Here “recurrent class = attractor” is legitimate and the gap (100 vs. 10) is genuine *cross-class* coherence: a 10-dimensional decoherence-free subspace built on the graph’s own pointer states.
- **Graph-unfaithful** (38 of 240): e.g. rule 22 — fixed-point operators carry weight 0.49 *outside* the recurrent block and $\Phi^{400}(\mathbb{K}/d)$ retains 12.5% of its population on graph-“transient” states (limit support: 11 basis states, not 3). Since $\dim \text{Fix} = 25 = 5^2$, the true attractor is a **5-dimensional decoherence-free subspace**; the “three point attractors” of the graph is an artifact of dephasing. Rule 50 ($\dim \text{Fix} = 13$, 25% leaked) and rule 106 (42%) are stronger cases.

For unfaithful rules the graph-level attractor language of §2 describes the *dephased* dynamics only; the coherent attractor requires Tier 1b. This is the quantitative form of the context’s warning that the graph under-resolves dissipative attractor structure.

W	tuple	#rec. states	$\dim \text{Fix}(\Phi)$	leaked weight	graph faithful?
0	DDDD	1	1	0.000	yes
200	DIII	10	100	0.000	yes
232	DIIE	6	36	0.000	yes
28	IIVD	3	9	0.000	yes
76	IIID	11	121	0.000	yes
22	IVVD	3	25	0.125	no
50	DVVV	1	13	0.250	no
178	DVVE	2	9	0.063	no
106	DEIV	1	8	0.422	no
census at $N = 4$, pbc: 202 faithful / 38 unfaithful of 240 dissipative rules					

Table 3: Exact faithfulness test (pbc, $N = 4$): leaked weight is the population of $\Phi^{400}(\mathbb{K}/d)$ remaining on graph-transient states. Non-zero leak means the graph misplaces the coherent attractor.

W	tuple	classical dim	Cesaro rank	coherence gap
0	DDDD	1	1	0
1	VDDD	15	15	0
22	IVVD	3	25	22
28	IIVD	3	9	6
50	DVVV	1	13	12
178	DVVE	2	9	7
200	DIII	10	100	90
232	DIIE	6	36	30

Table 4: Exact Cesaro rank vs. the graph-resolved per-class fixed dimension (pbc, $N = 4$). A zero gap (rule 0, pure decay) is fully classical; a large gap flags protected coherence. Read together with Table 3: for faithful rules the gap is cross-class coherence, for unfaithful rules it also includes structure the graph places on the wrong states.

5 Transient relaxation

Figure 1 shows the transient depth (an exact integer proxy for the relaxation time) growing with N for representative multistable rules.

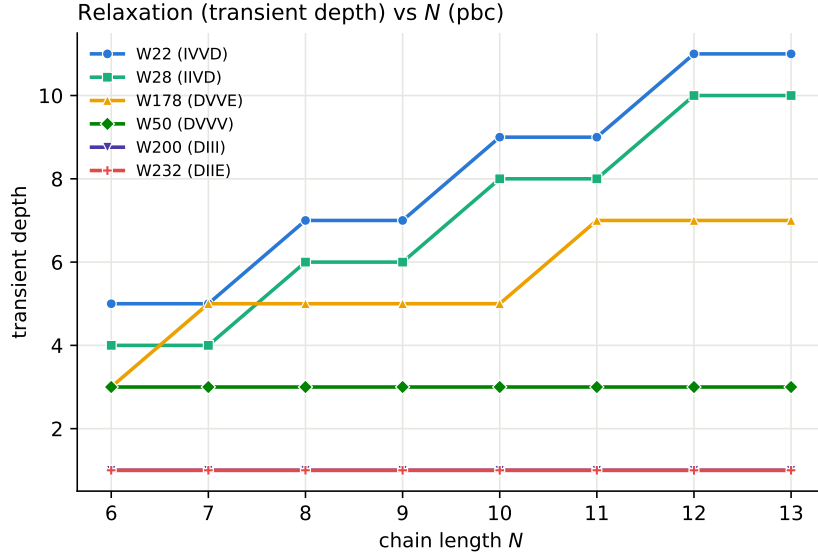


Figure 1: Transient depth vs. N (pbc) for selected dissipative rules.

6 Status and next steps

The graph-level dissipative classification is complete and checkpointed; Tier-1b per-class attractor types are exact and validated. The protected-coherence gap identifies the rules whose full attractor algebra (cross-class noiseless subsystems, weak-symmetry coherences) exceeds the graph's resolution — the targets for a complete Tier-1b/Tier-2 treatment. All numbers derive from `results/*.jsonl` (engine_version 1a.1) and the exact superoperator engine; regenerated by `python -m qca_fragmentation.quantum.dissipative_report`.